

You are working on the **public release of BMFinder**, a Chromium Manifest V3 browser extension for Brave, Chrome, and Edge.

Your task is to audit and refine the public build so its language, information architecture, defaults, privacy disclosures, exports, and deletion controls accurately communicate its primary purpose:

BMFinder helps Arma 3 players save, recognise, and reconnect with friends, teammates, and familiar community members through BattleMetrics information already visible to the user.

This is an implementation task, not merely a design proposal. Inspect the existing repository, understand the current architecture and state flow, make the changes described below, update tests and documentation, and report exactly what changed.

Do not rewrite the extension from scratch. Preserve reliable existing functionality, the Material 3 token system, current theme support, IndexedDB architecture, extraction centralisation, and the existing deliberate pacing of page navigation.

1. Product context

Arma 3 players frequently change their in-game names. The game does not provide a convenient way to connect an in-game identity with a Steam friend identity. Adding someone on Steam normally requires both people to exchange an exact name or profile link, which can be mistyped or difficult to find.

BMFinder exists primarily to let a user:

- Save friends and familiar players.
- Give them a persistent personal label.
- Recognise them after they change their in-game name.
- See whether they are currently playing.
- See which followed server they were last observed on.
- Find other familiar people they may want to join.
- Maintain a short, private, local history.
- Transfer a deliberately selected list to a trusted friend where appropriate.

BMFinder is not intended to operate as:

- A public player-intelligence database.
- A stalking or harassment tool.
- A method for bypassing private BattleMetrics profiles.
- An automated accusation or enforcement system.
- A commercial player-data product.
- A way to infer a hidden person's identity from missing roster entries.

2. Public-build invariants

Treat the following as mandatory safety and release requirements.

2.1 Features that must not exist in the public build

The public build must not contain, expose, register, route to, or silently retain:

- Incremental player-ID range scanning.
- Any control that walks consecutive player IDs.
- Absence correlation.
- Admin-versus-player presence correlation.
- Suspicion scoring.
- Behavioural risk scoring.
- Automatic recommendations to investigate, punish, flag, or ban a player.
- A “Suspect” relationship role.
- UI language that claims an unavailable player is a hidden administrator.
- Logic that attempts alternative endpoints specifically to reveal a profile after BattleMetrics returns it as private, unavailable, 404, or not found.

Search the complete public build—not only visible components—for old routes, handlers, strings, storage keys, commands, tests, feature flags, CSS, documentation, and dead imports related to these capabilities.

Remove them from the compiled public extension rather than merely hiding buttons with CSS.

Do not remove anything from a clearly separate private development branch or private edition unless the repository structure requires separation to prevent it being bundled publicly. Where necessary, create a clean build-time boundary so the public package cannot accidentally include the excluded code.

2.2 Privacy controls must be respected

BattleMetrics may return a profile as private, unavailable, deleted, inaccessible, or not found.

When this occurs:

- Stop the lookup normally.
- Do not try to infer the player’s identity.
- Do not label the player as an administrator.
- Do not claim the user has deliberately hidden themselves.
- Do not imply wrongdoing.
- Do not attempt alternate routes intended to defeat the unavailable state.
- Do not expose raw error details that encourage circumvention.

Use neutral copy:

Title: “Profile unavailable”

Body:

“BattleMetrics is not making this profile available to your current session. It may be private, deleted, temporarily unavailable, or inaccessible with your current permissions.”

Where an already saved player becomes unavailable, preserve the user's own local nickname and tags, but clearly distinguish them from current BattleMetrics information.

Display:

- User label: preserved.
- Last successfully observed in-game name: marked as historical.
- Current status: unavailable.
- Last successful refresh time.
- A "Remove saved person" action.
- A "Delete this person's history" action.

Do not continuously retry unavailable profiles. Apply a sensible backoff and allow an intentional manual retry.

3. Information architecture and terminology

Move the public interface away from investigative or surveillance terminology and toward friendly, neutral social language.

Do not make the interface childish or overly cheerful. Keep it mature, precise, and suitable for an Arma community.

3.1 Navigation terminology

Use the following preferred navigation model, adapting only where the existing layout makes a close equivalent substantially clearer:

- **Search** → **Find players**
- **Watchlist** → **People**
- **Servers** → **Servers**
- **Monitor** → **Live activity**
- **Archive** → **Recent activity**

The "People" tab should be described as:

"Friends, teammates, and familiar players you have chosen to save."

The "Live activity" tab should answer:

"Is anyone I know playing right now?"

The "Recent activity" tab should answer:

"Who did BMFinder observe during recent server checks?"

3.2 Action terminology

Prefer:

- “Add to watchlist” → “Save person”
- “Remove from watchlist” → “Remove saved person”
- “Refresh watchlist” → “Refresh saved people”
- “Poll servers” → “Check servers now”
- “Continuous polling” → “Live updates”
- “Start monitoring” → “Start live updates”
- “Stop monitoring” → “Stop live updates”
- “Tracked player” → “Saved person” or “Saved player”
- “Tracking” → “Saved” or “Following,” depending on context
- “Nickname” → “Your label”
- “Role” → “Relationship”
- “Last seen” → “Last observed playing”
- “Archive snapshot” → “Server check”
- “Presence history” → “Recent observations”

Use “observe” rather than “detect” where technically accurate.

Avoid the terms:

- Suspect.
- Target.
- Intelligence.
- Hidden admin.
- Surveillance.
- Investigation candidate.
- Risk score.
- Correlation subject.
- Evidence.

Technical documentation may use accurate engineering terminology where necessary, but user-facing language must reflect the social purpose.

3.3 Relationship categories

Replace the current public default roles with:

- Friend
- Teammate
- Community member
- Server staff
- Other

Do not automatically assign any category that implies suspicion.

Rename the field from “Role” to “Relationship.”

Existing stored values must be migrated safely:

- `player` → `Community member`
- `admin` → `Server staff`
- `other` → `Other`
- `suspect` → `Other`

Do not silently lose user records during migration.

Where an existing `suspect` value is migrated, optionally show a one-time notice:

“An older relationship label was replaced with ‘Other’ because the public version now uses neutral social categories.”

Do not include accusatory wording in that notice.

Custom tags remain user-defined. Do not police ordinary custom labels locally, but do not ship predefined accusatory tags.

4. First-run disclosure and consent

Create a polished first-run disclosure shown before BMFinder begins passive recording or server checks.

It must be concise enough to read, but specific enough to explain the actual behaviour.

4.1 First-run screen

Heading:

“Welcome to BMFinder”

Intro:

“BMFinder helps you save and recognise friends, teammates, and familiar Arma 3 players using BattleMetrics pages available in your browser.”

Include four clear disclosure cards or sections:

Stored locally

“Saved people, labels, server information, and recent observations are stored in this browser profile. BMFinder does not send them to a BMFinder server.”

What BMFinder reads

“When you visit supported BattleMetrics pages, BMFinder can read relevant player and server information needed to update people and servers you have chosen to save.”

Intentional server checks

“When you choose a refresh or enable live updates, BMFinder opens BattleMetrics pages sequentially using your existing browser session. Checks are deliberately paced.”

Private profiles

“BMFinder respects profiles that BattleMetrics makes private or unavailable and does not attempt to reveal them.”

Include links or buttons for:

- Read Privacy Policy
- Read Acceptable Use Policy
- View data controls

Require an affirmative button:

“I understand — continue”

Do not use a preselected checkbox.

Persist the accepted policy version and date. If data practices materially change, show a new disclosure rather than silently treating prior acceptance as sufficient.

No server polling or passive persistence should occur before the first-run disclosure has been accepted, except for the minimum storage required to remember that the disclosure has not yet been completed.

5. Contextual privacy explanations

Add subtle explanations where users make privacy-relevant choices.

5.1 People tab empty state

Use:

Heading: “Save people you recognise”

Body:

“Add friends, teammates, or familiar players so BMFinder can help you recognise name changes and see when they appear on servers you follow.”

Primary action:

“Find players”

Secondary action:

“Import saved links”

Do not use imagery or language resembling a law-enforcement watchlist.

5.2 Live updates explanation

Before first enabling live updates, show:

Heading: “Enable live updates?”

Body:

“BMFinder will check your followed servers one at a time while this dashboard remains open. Recent observations are stored locally for [RETENTION PERIOD]. You can pause updates or delete the history at any time.”

Actions:

- Enable live updates
- Not now

Include a link to retention settings.

5.3 Observation limitations

Near “Last observed playing” and in Recent activity, provide a tooltip or info panel:

“BMFinder only knows about checks that occurred while it was running. A missing observation does not mean someone was offline.”

Where appropriate, add:

“Private sessions, hidden server rosters, delayed information, name changes, or failed checks can make observations incomplete.”

Never visually present missing information as a suspicious event.

6. Export redesign

Exports are useful for backups and deliberately sharing a social list, but player identifiers and history should not be included casually.

Replace any one-click CSV export with an explicit export dialog.

6.1 Export dialog

Heading: “Export BMFinder data”

First ask for the purpose:

- Personal backup

- Share a saved-people list
- Export recent activity
- Custom export

Then provide granular checkboxes.

Saved-person fields

- Your labels — default on
- Current in-game names — default on
- Relationship categories — default on
- Tags — default on
- Player IDs — default off
- Last observed server — default off
- Last observation time — default off

Server fields

- Server names — default on where relevant
- Server links — default on
- Server IDs — default off

Activity fields

- Recent server checks — default off
- Unsaved roster members — default off and visually marked sensitive
- Exact timestamps — default off
- Player IDs in activity history — default off

Do not include unsaved roster members in a “Share a saved-people list” export.

6.2 Export warning

When any of the following are enabled:

- Player IDs.
- Exact activity timestamps.
- Unsaved roster members.
- Full server-check history.

Show:

Heading: “This export contains detailed player information”

Body:

“Only share it with someone you trust and only for a legitimate social, backup, or authorised community purpose. Do not publish it as a public player database or use it for harassment, identity tracing, or unwanted monitoring.”

Require a confirmation checkbox that is not preselected:

“I have reviewed the selected fields.”

6.3 Export formats

Support:

- A native BMFinder transfer format for importing into another BMFinder installation.
- CSV where tabular interoperability is useful.
- JSON backup for complete personal restoration.

Clearly distinguish them:

- **BMFinder list:** intended for transferring selected saved people.
- **CSV:** intended for manual review or use in spreadsheet software.
- **Full backup:** intended for restoring the same user's data.

Do not make a full backup the default sharing format.

6.4 Import safeguards

When implementing BMFinder-list or CSV import:

- Preview the number of records.
- Show which fields will be imported.
- Deduplicate by stable player ID where available.
- Never overwrite a user label without confirmation.
- Do not import activity history by default.
- Do not import unsaved roster snapshots through a friends-list workflow.
- Allow the user to choose a default relationship category.
- Clearly identify malformed or unsupported rows.
- Keep the import local.

7. Deletion and retention controls

Create a dedicated **Data and privacy** settings section.

It must show:

- Number of saved people.
- Number of followed servers.
- Number of recent server checks.
- Oldest retained observation.
- Current retention period.
- Approximate local storage use where reasonably available.

Provide these controls:

- Delete an individual person's history.
- Delete an individual server's history.
- Clear recent activity.
- Remove all saved people.
- Remove all followed servers.

- Reset labels and tags.
- Delete all BMFinder data.
- Export backup before deletion.

7.1 Retention settings

Provide retention choices such as:

- 1 day
- 3 days
- 7 days
- 14 days
- 30 days
- Custom, capped at a documented maximum

Keep the public default at 14 days unless a lower default is already required by product decisions.

Explain:

“Retention applies to recent activity and server-check history. Saved people, labels, tags, and followed servers remain until you remove them.”

When the user shortens retention, prune older data immediately after confirmation.

7.2 Delete-all flow

Use a deliberate confirmation:

Heading: “Delete all BMFinder data?”

Body:

“This permanently removes saved people, labels, tags, followed servers, recent activity, and settings from this browser profile. Exported files will not be deleted.”

Require typing:

DELETE

Do not use a vague “Are you sure?” dialog.

8. Privacy blur

Preserve the existing screenshot privacy-blur feature and improve its explanation.

Rename it if needed to:

“Screenshot privacy”

Tooltip:

“Blur in-game names and player IDs before sharing a screenshot. Hovering may temporarily reveal a field on your own screen.”

Add an optional stronger mode:

“Presentation mode”

In presentation mode:

- Names remain blurred even on hover.
- Player IDs remain blurred.
- Export buttons are hidden or disabled.
- Tooltips do not reveal identifiers.
- The state is clearly indicated.
- Leaving presentation mode requires an intentional action.

Do not imply that screenshot blur removes data from storage. It affects only presentation.

9. About, independence, and policy pages

Add an About or Legal section accessible from the dashboard and popup.

It must state:

“BMFinder is an independent community project. It is not affiliated with, endorsed by, sponsored by, or produced by BattleMetrics, Valve, Steam, Bohemia Interactive, or Arma.”

Include accessible pages for:

- Privacy Policy.
- Terms of Use.
- Acceptable Use Policy.
- Open-source licence, if applicable.
- Contact and security reporting.
- Current version.
- Data-schema version.
- Extraction version.

Do not use BattleMetrics logos, marks, or visual branding in a manner that suggests endorsement.

Use plain text when referring to BattleMetrics unless an authorised asset is available.

10. Store-listing single purpose

Prepare suggested Chrome Web Store copy that reflects one clear purpose.

Short description

"Save and recognise friends and familiar Arma 3 players through BattleMetrics, with local labels, server activity, and name-change awareness."

Single-purpose statement

"BMFinder helps Arma 3 players save, recognise, and reconnect with friends, teammates, and familiar community members using player and server information displayed on BattleMetrics pages available in their browser."

Privacy summary

"BMFinder stores its working data locally in your browser profile. It does not operate a developer-controlled player database or sell player information."

Do not market the public extension using:

- Hidden-player discovery.
- Admin detection.
- Cheater identification.
- Behavioural correlation.
- Investigation.
- Intelligence gathering.
- Surveillance.
- Player-ID scanning.

11. Technical constraints

Preserve the existing architectural strengths:

- Chromium Manifest V3.
- One background worker as the controlled storage writer where currently designed.
- Centralised DOM selectors and extraction logic.
- Sequential hidden-tab reuse.
- Deliberate gaps between page loads.
- IndexedDB for structured local storage.
- Existing backup compatibility where safe.
- Theme tokens and light/dark/system support.
- Bundled fonts with no tracking CDN.
- Visible extraction failure warnings.

Do not introduce:

- A remote analytics service.
- Developer-accessible telemetry.
- A cloud player database.
- Background transmission of saved-player data.
- External font or tracking requests.

- Fingerprinting.
- Undisclosed permissions.
- Broad host permissions unrelated to BattleMetrics.
- Automatic high-volume retries.

Audit `manifest.json` and request only the narrowest permissions necessary for current public features.

Where the bookmarks permission is optional, request it only when the user initiates browser-bookmark import rather than requiring it for unrelated functionality, where Chromium's permission model allows this.

12. Storage and migration

Inspect the current IndexedDB schema and implement safe migrations.

Requirements:

- Existing saved players must survive terminology changes.
- Existing labels and tags must survive.
- Existing servers must survive.
- Existing activity history must remain subject to retention.
- Removed public features must not leave active scheduled jobs.
- Removed public features must not leave visible data that implies suspicion.
- Migration must be idempotent.
- Migration failure must not silently erase the database.
- Create a backup or transactional rollback strategy before destructive schema changes where practical.
- Record the schema version.

Add tests for:

- Role-to-relationship migration.
- `suspect` to `Other` migration.
- Unavailable/private profile state.
- Export defaults.
- Exclusion of player IDs by default.
- Deletion of per-player history.
- Deletion of per-server history.
- Retention pruning.
- Removal of obsolete scheduled jobs.
- First-run disclosure versioning.
- Public build exclusion of scanning and correlation modules.

13. Accessibility and UX quality

Maintain or improve:

- Keyboard navigation.

- Visible focus states.
- Semantic controls.
- Screen-reader labels.
- Sufficient contrast in all accent themes.
- Reduced-motion support.
- Clear loading and failure states.
- Confirmation messages that explain what happened.
- Non-colour indicators for relationship categories and statuses.

Do not use colour alone to distinguish privacy, failure, activity, or relationship state.

Keep dense information readable at common Chromium zoom levels, including 90% and 100%.

Avoid overly small metadata text.

14. Empty, loading, and error states

Design states for:

- No saved people.
- No followed servers.
- No recent activity.
- A server that has never been checked.
- A person never observed by BMFinder.
- A player profile that is unavailable.
- A BattleMetrics page layout change.
- A timed-out check.
- A paused live-update session.
- An empty import.
- A duplicate import.
- A partially successful export.
- Storage quota issues.

Error messages must not imply malice or privacy evasion.

Example extraction error:

“BMFinder could not read the expected information from this BattleMetrics page. The page layout may have changed. No record was updated.”

15. README changes

Rewrite the public README so the opening immediately establishes the social purpose.

Suggested opening:

BMFinder

BMFinder is a local browser extension that helps Arma 3 players save and recognise friends, teammates, and familiar community members through BattleMetrics.

Players often change their in-game names, and Arma 3 does not provide a simple way to connect an in-game identity with a Steam friend. BMFinder lets you keep your own labels, follow familiar players and servers, and check whether anyone you know is playing.

BMFinder stores its working data in your browser profile. It does not operate a central player database and does not attempt to reveal profiles that BattleMetrics makes private or unavailable.

Retain useful technical explanations, but revise or remove passages that make the public extension sound like an anti-bot workaround, hidden-admin finder, scanner, or investigative tool.

It is acceptable to explain honestly that the extension reads rendered pages instead of reproducing private request-signing logic. Phrase this as a technical boundary, not an achievement in defeating protections.

Move detailed historical experiments involving API failures and Cloudflare challenges into a development-history document if they are not necessary for ordinary installation or use.

Ensure the public README accurately states that:

- ID-range scanning is not included.
- Absence correlation is not included.
- Private and unavailable profiles are respected.
- Data is local unless deliberately exported.
- Observations are incomplete and are not proof.
- Live updates are deliberately paced.
- The project is independent and not endorsed by BattleMetrics.

16. Acceptance criteria

The work is complete only when all of the following are true:

1. The visible public interface contains no "Suspect" role or accusatory defaults.
2. The public build contains no ID-range scanner.
3. The public build contains no absence-correlation engine or UI.
4. Private or unavailable profiles receive a neutral, non-inferential state.
5. The interface does not identify roster omissions as hidden administrators.
6. The primary user journey is framed around friends, teammates, and familiar players.
7. A first-run disclosure appears before data recording begins.
8. Privacy, Terms, and Acceptable Use pages are accessible in-product.
9. Exported player IDs are off by default.
10. Unsaved roster members are off by default in every export.
11. Sharing exports require deliberate field selection.
12. Per-player, per-server, recent-history, and delete-all controls exist.

13. Retention is visible and configurable.
14. Existing user records migrate without loss.
15. Manifest permissions are audited and justified.
16. Tests cover migrations, retention, deletion, export defaults, and unavailable profiles.
17. The README and store copy accurately match the public build.
18. No remote analytics, telemetry, or developer-controlled player database has been introduced.
19. Existing light, dark, system, accent, typography, and privacy-blur functionality remains working.
20. `npm test` passes, with new tests added where appropriate.

17. Required working process

Begin by inspecting:

- `manifest.json`
- Background worker and job queue.
- Content reader.
- DOM extraction library.
- IndexedDB schema and migrations.
- Correlation and matching modules.
- Dashboard routes and tabs.
- Popup actions.
- Export and backup code.
- Existing tests.
- README and any store metadata.
- Build or packaging scripts.

Then:

1. Identify every public-build location affected by this prompt.
2. Make the implementation changes.
3. Add or update migrations.
4. Add or update tests.
5. Run the full test suite.
6. Build or package the public extension if a build command exists.
7. Inspect the generated package to confirm excluded features are not bundled.
8. Report any requirements that could not be completed and explain why.

Do not merely produce mockups or pseudocode where the repository allows a real implementation.

18. Final report format

At completion, provide:

Summary

A concise description of the new product framing and safeguards.

Files changed

Each modified file with a one-sentence explanation.

Public features removed

Confirm exactly how ID scanning, absence correlation, suspicious roles, and related code were excluded.

UX changes

Explain navigation, terminology, first-run disclosure, privacy states, exports, and deletion controls.

Data migration

Explain how existing records and old role values are migrated.

Privacy and permissions

List the manifest permissions, why each is required, and whether any were removed or made optional.

Tests

List tests added or changed and provide the final test result.

Build verification

Confirm whether excluded functionality appears anywhere in the generated public package.

Remaining risks

State any unresolved privacy, compatibility, legal, or BattleMetrics-policy questions without pretending they have been resolved.